

SAiDL Induction Assignment 2026

by AB

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1 Introduction

1.1 Assignment

- This Report attempts to tackle the Core-ML and Diffusion problem statement from the SAiDL Induction 2026 Assignment
- All evaluation is done on a NVIDIA GeForce RTX 3050 6GB Laptop GPU, so all the results pertaining to GPU analysis, throughput etc are in accordance with aforementioned device
- GitHub Repo- https://github.com/Ad-B-R/SAiDL_Induction_Assignment

1.2 Assignment-1

- Core-ML Assignment tackles the empirical study of different attention variants and their performance.
- For this assignment, Sliding Window, Grouped Query Attention and SoftMax-Free Attention were used alongside the baseline Attention model
- For Positional Embeddings- Standard sine-cosine absolute encoding, ALiBi, RoPE and RPE were used
- Lastly, the Convolution + Attention model uses (insert model) based on the empirical results from the 2 and 3 subsection of the assignment

1.3 Assignment-2

- Core-ML Assignment tackles the empirical study of different attention variants and their performance.
- For this assignment, Sliding Window, Grouped Query Attention and SoftMax-Free Attention were used alongside the baseline Attention model
- For Positional Embeddings- Standard sine-cosine absolute encoding, ALiBi, RoPE and RPE were used
- Lastly, the Convolution + Attention model uses (insert model) based on the empirical results from the 2 and 3 subsection of the assignment

2 Core ML Assignment

2.1 Introduction

- As was discussed earlier, this assignment is the empirical study of different attention variants and their performance, incorporating ablation study of the context window etc.

2.2 Baseline Attention

2.2.1 Learnings from the paper

- Architecture behind transformer (the FFN, use of Residual connection in conjunction with the Add+Norm Layer) specifically the Decoder-Part
- Casual masking, which retains the auto-regressive property that is needed to make the architecture for wikitext-2 dataset, whose implementation mimics GPT's (Generative Pre-Trained Transformer) work
- It also described the absolute sin-cosine positional encoding (which we will call "standard" encoding from now on)
- The attention formula itself which is the basis of the "attention" block present in the architecture (and would be the point of introduction of RoPE and ALiBi)

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

- Need of two separate embeddings- Positional Embedding and Word Embedding, whereby one dictates the Position of individual sequence's order, crucial for NLP tasks; the other one to actually give words meaning in a vector form

2.2.2 Implementation Assumptions

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2.2.3 Implementation Learnings

2.3 Different Attention Mechanism

2.3.1 Learnings from the paper

- a. Sliding Window Attention
- b. Grouped Query Attention
- c. SoftMax-Free Attention

2.3.2 Implementation Assumptions

2.3.3 Implementation Learnings

2.4 Experiment Analysis

2.4.1 Graphs

- All the following graphs are evaluated on Absolute Standard Positional encoding, ranging from context length $L_{train} = [256, 512, 768, 1024]$
- The Attention Mechanism tested are- Grouped Query Attention (GQA), SoftMax-Free Attention (ReLU-Attention) and Sliding Window Attention

Analysis for $L_{train} = 256$

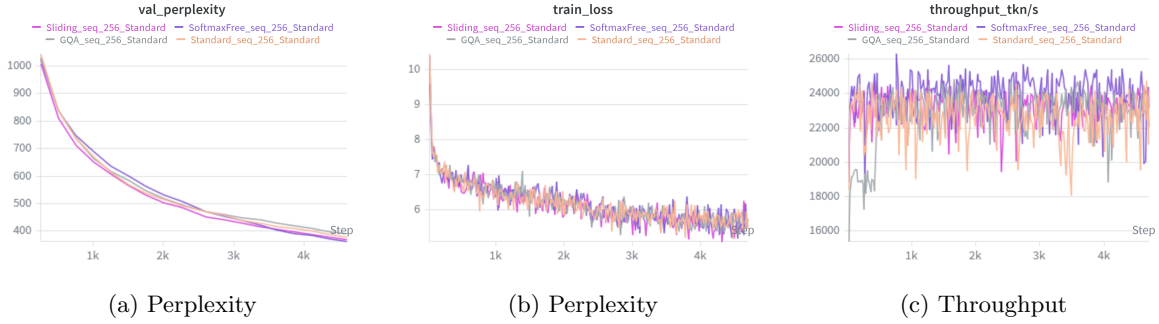


Figure 1: Results from $L_{train} = 256$ indicate SoftMax-Free has the highest throughput meanwhile having the lowest perplexity

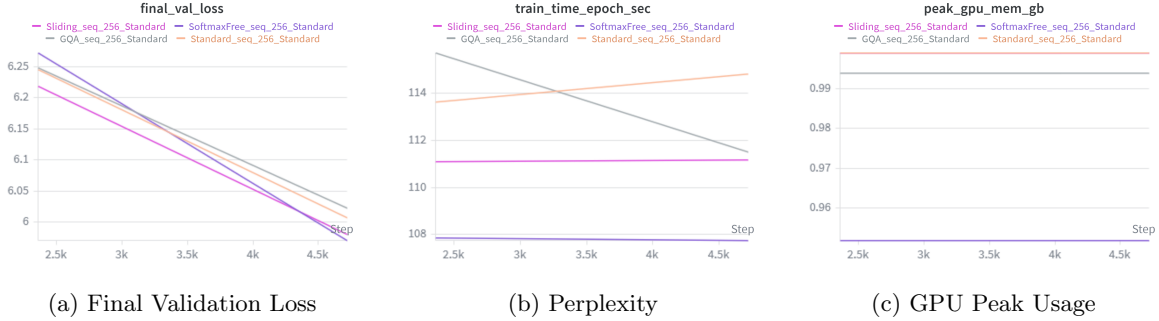


Figure 2: Results from $L_{train} = 256$ indicate SoftMax-Free has the least GPU mem usage meanwhile having the highest negative final validation loss slope

- It is imperative to note that while SoftMax-Free performed the best in terms of validation loss and perplexity (all other graphs relevant to this are attached in Subsection 2.7 which might show a slight variance but this is the trend for the major important graphs), Sliding Window Consistently performed the 2nd best for $L_{train} = 256$, a fact which will become relevant for the other sequence length further. It can also be noted the GQA had the highest perplexity, val loss and throughput, while it had the 3rd highest GPU peak Usage, hence making it the worst performer in this bunch.

Analysis for $L_{train} = 512$

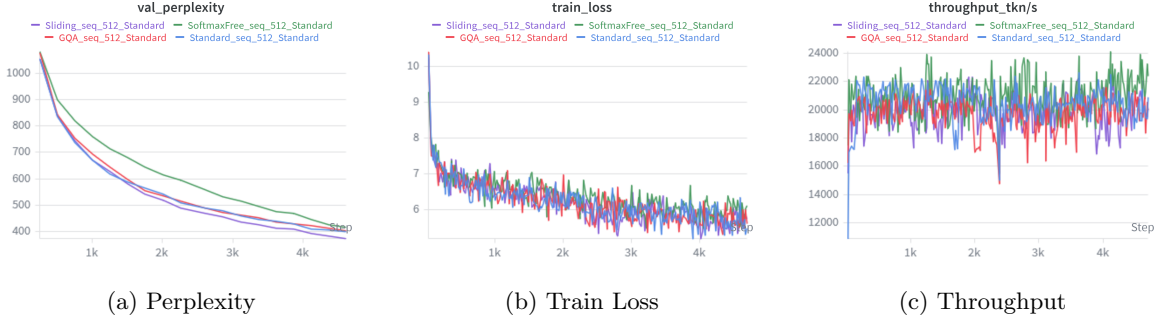


Figure 3: Results from $L_{train} = 512$ indicate SoftMax-Free has the highest throughput while having the highest perplexity meanwhile Sliding Window has the lowest perplexity

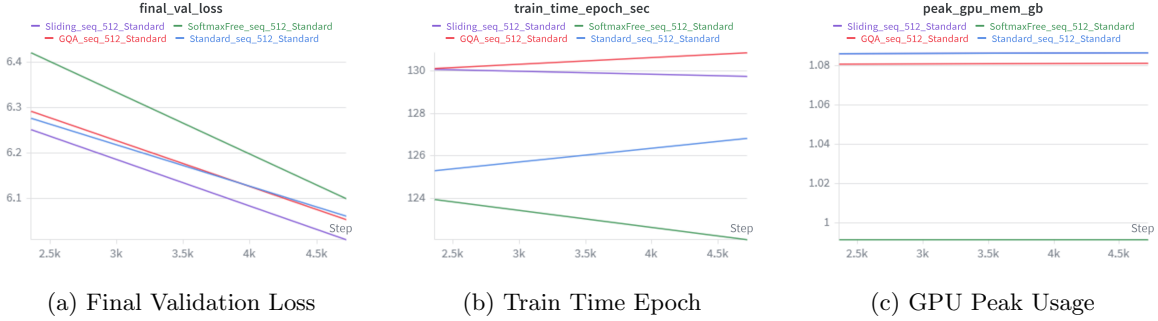


Figure 4: Results from $L_{train} = 512$ indicate SoftMax-Free has the least GPU mem usage meanwhile having the highest final validation loss (and slope too) while Sliding Window has the lowest final validation loss while having same peak mem usage as Standard

- It is imperative to note that SoftMax-Free performed the best in terms of peak gpu memory usage and the throughput, it performed drastically worse on perplexity and val loss (having the highest by a long margin compared to the others). This would contribute to some interesting inferences as will be discussed later.
- Sliding Window consistently performed the best on perplexity/loss front by having the least perplexity/loss by far (Although a high peak gpu memory usage and the least throughput).
- Standard and GQA showed identical trends in all metrics
- For $L_{train} = 768, 1024$, a similar trend was observed as $L_{train} = 512$

Analysis for $L_{train} = 768$

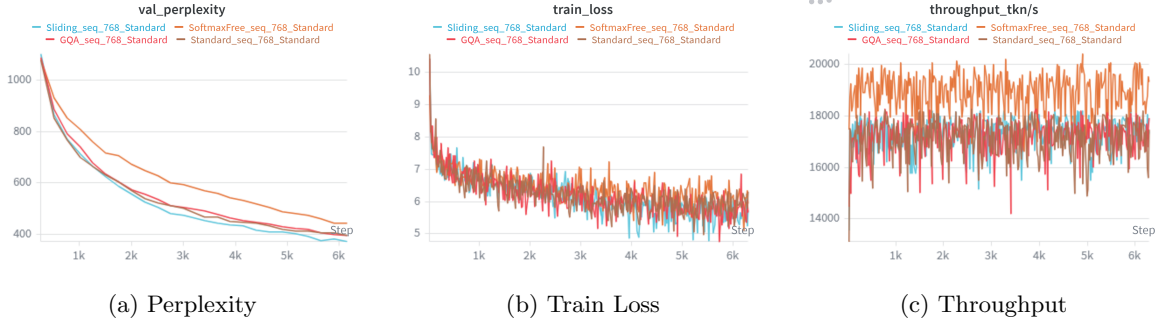


Figure 5: Results from $L_{train} = 768$ indicate SoftMax-Free has the highest throughput while having the highest perplexity meanwhile Sliding Window has the lowest perplexity

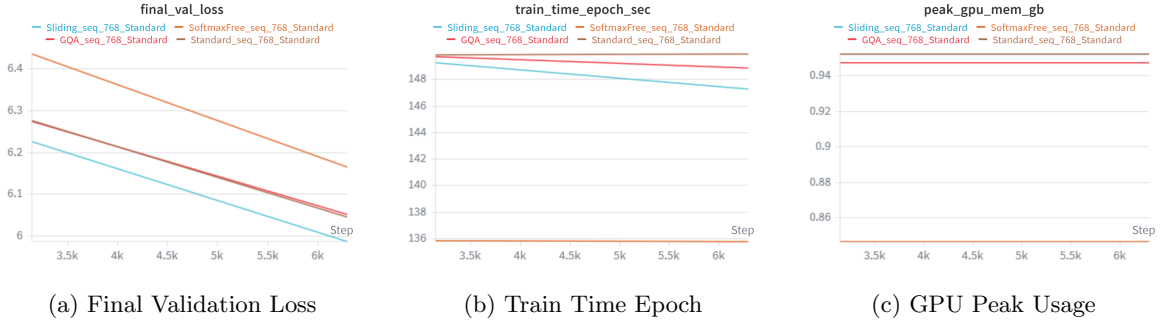


Figure 6: Results from $L_{train} = 768$ indicate SoftMax-Free has the least GPU mem usage meanwhile having the highest final validation loss (and slope too) while Sliding Window has the lowest final validation loss while having same peak mem usage as Standard

Analysis for $L_{train} = 1024$

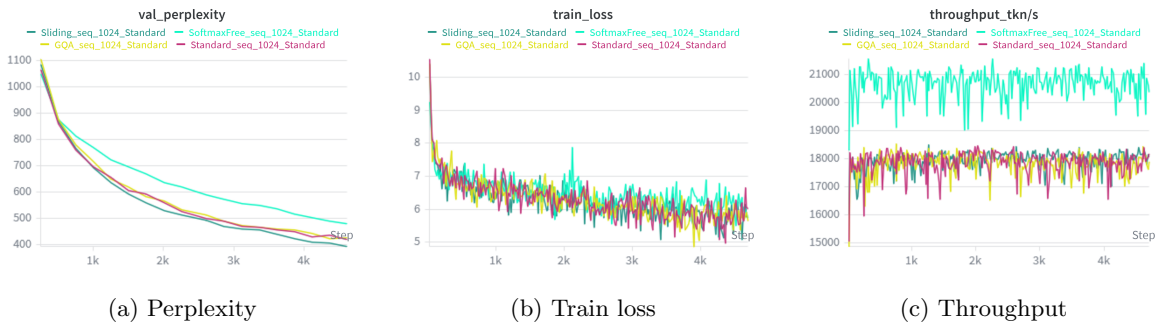


Figure 7: Results from $L_{train} = 1024$ indicate SoftMax-Free has the highest throughput while having the highest perplexity meanwhile Sliding Window has the lowest perplexity

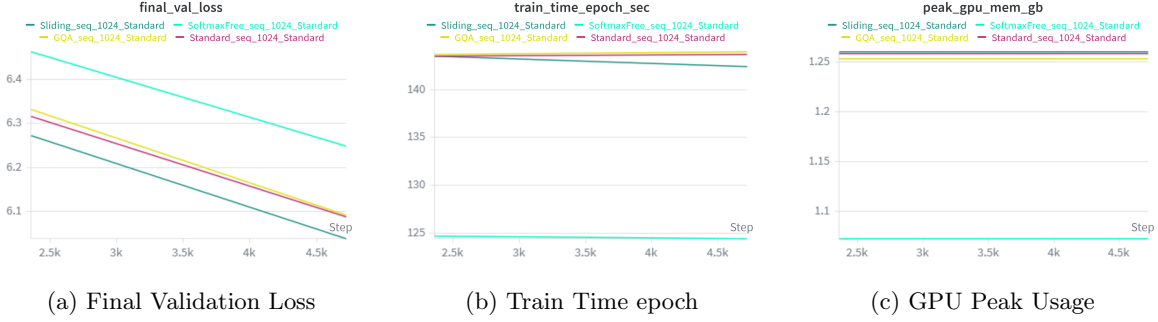


Figure 8: Results from $L_{train} = 1024$ indicate SoftMax-Free has the least GPU mem usage meanwhile having the highest final validation loss (and slope too) while Sliding Window has the lowest final validation loss while having same peak mem usage as Standard

2.4.2 Inference

- Sliding Window consistently performed the best in terms of validation loss/perplexity (minimal) (across all given sequence length), hence it would be the choice as the best model for the following tests.
- SoftMax-Free performed the best in terms of throughput, Peak GPU mem usage (least), avg time per epoch (least) and so on, hence it is the fastest of the 4 choices. At the same time it had the worst loss/perplexity (barring $L_{train} = 256$) amongst the given attention mechanisms
- Across Validation Loss and perplexity tests, GQA and Standard Attention performed similarly for varying sequence length, hence there is not much difference in the performance of the two.
- GQA becomes faster (lower peak gpu usage, avg time per epoch etc barring $L_{train} = 512$ where GQA is more expensive than Standard attention) than Sliding Window and Standard Attention with an increase in sequence length, so it gives the same performance as standard attention at less overhead

2.4.3 Interpretation

- Sliding Window being the best could be attributed to the inherent inductive bias associated with human language which Standard Attention, which by extension applies to SoftMax-Free and GQA attention mechanisms.
- SoftMax-Free being the fastest could be attributed to how that specific model uses ReLU, which doesn't actually employ the use of any exponentials, which is a more expensive operation for CUDA to handle, hence the better performance

2.5 Different Positional Encoding Implementation

2.5.1 Learnings from the paper

2.5.2 Implementation Assumptions

2.5.3 Implementation Learnings

2.6 Experiment Analysis

2.7 Convolution Attention Implementation

2.7.1 Empirical Results From Previous Runs

From the experiments done previously, it is clear that Sliding Window is the clear choice for Attention Mechanism and ALiBi is the clear choice for Positional Encoding. Using this empirical result, we will be working on the Convolution + Attention architecture

3 Diffusion

3.1 Positional Encoding Ablation

3.2 Throughput Analysis

3.3 Downstream Perplexity